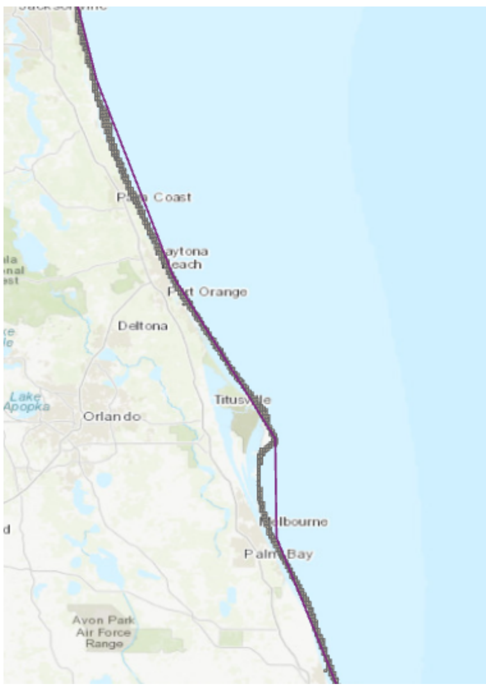
Amelia Keefe: Lab 2

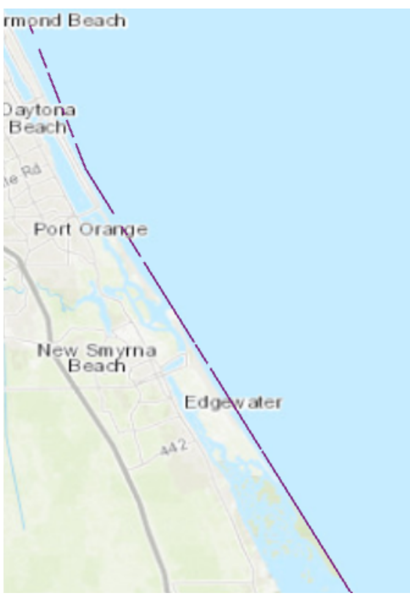
- What information (attributes) is included?
  - The information included is FID, shape, scale rank, feature class, and minimum zoom.
- How does vector data represent geographic features?
  - Vector data represents geographic features as geometric types such as points, lines, and polygons.
- What types of spatial features can you identify in the map?
  - I can identify line features in my map.
- How does raster data differ from vector data?
  - Raster data represents continuous fields and comprises grid cells, while vector data represents discrete objects using geometric shapes such as points, lines, and polygons.
- What information does the DEM provide?
  - The Digital Elevation Model (DEM) provides elevation data that can be used to estimate slope and terrain features.
- Why is clipping important in GIS analysis?
  - Clipping is important because it allows researchers to focus on specific areas of interest by extracting only the data within a defined boundary.
- What are potential real-world applications of raster analysis?
  - Raster analysis can be applied in remote sensing, such as training an AI to differentiate between native and invasive tree species in a specific region.

Reflection:

Vector and raster data combine detailed geometric representations of specific features with continuous surface information. For example, vector layers like administrative boundaries can be overlaid with raster datasets like land cover or elevation to create a complete picture. This approach is useful in mapping flood-prone areas to identify at-risk neighborhoods and plan evacuation routes, improving disaster response and urban planning efforts.


Palm Coast
Daytona Beach
Port Orange
Deltona
Lake Apopka
Orlando
Titusville
Melbourne
Palm Bay
Avon Park Air Force Range

Clip:


Ormond Beach
Daytona Beach
Port Orange
New Smyrna Beach
Edgewater
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